

ESE 101 Homework 4

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1 Pasadena Air Pollution

Air pollution is a serious problem in Los Angeles. In addition to well known health problems, like increased cancer and asthma rates, the smog can also darken the skies by scattering incoming shortwave radiation. Not only is `aq.caltech.edu` a good site to check the current air quality, but it also lists live concentrations of two common shortwave scatters, PM2.5 and PM10 taken on top of Caltech tower. PM2.5 indicates the concentration of particulate matter below 2.5 microns (10^{-6}m) while PM10 indicates the concentration below 10 microns (which includes PM2.5). In this question we will compute the optical depth and estimate the affect of these two particles on radiation reaching the surface. We'll assume a gray atmosphere and ignore effects on shortwave from other absorbers and scatterers.

Part 1

Visit `aq.caltech.edu` and tell us the time you checked and what the concentrations of PM2.5 and PM10 are. Compute the difference to estimate the fraction of absorbers greater than 2.5 microns. Please convert to SI units.

The following air quality data was checked on `November 6, 2025, at 2:10 PM` from `aq.caltech.edu`. The mass concentrations are converted from $\mu\text{g}/\text{m}^3$ to the SI unit of kg/m^3 .

- **PM2.5:** $16.3\mu\text{g}/\text{m}^3 = 1.63 \times 10^{-8} \text{ kg}/\text{m}^3$
- **PM10:** $29.8\mu\text{g}/\text{m}^3 = 2.98 \times 10^{-8} \text{ kg}/\text{m}^3$

The problem states that PM10 includes PM2.5. The mass concentration of particles in the range (2.5 μm , 10 μm] is the difference between these two values.

- **Difference (PM2.5-10):**

$$(29.8\mu\text{g}/\text{m}^3) - (16.3\mu\text{g}/\text{m}^3) = 13.5\mu\text{g}/\text{m}^3$$

In SI units, this is $1.35 \times 10^{-8} \text{ kg}/\text{m}^3$.

- **Fraction of particles > 2.5 microns:** This is the fraction of the total PM10 mass that is in the 2.5-10 micron range.

$$\text{Fraction} = \frac{\text{Concentration (PM}_{2.5-10})}{\text{Concentration (PM}_{10})} = \frac{13.5\mu\text{g}/\text{m}^3}{29.8\mu\text{g}/\text{m}^3} \approx 0.45302$$

Approximately 45.3% of the PM10 particulate mass is from particles larger than 2.5 microns.

Part 2

Assume the mass extinction coefficients, $k_e^{\text{PM}_{2.5}} = 4\text{m}^2\text{g}^{-1}$ and $k_e^{\text{PM}_{2.5-10}} = 1\text{m}^2\text{g}^{-1}$, compute the volume extinction coefficients, $\alpha_e^{\text{PM}_{2.5}}$ and $\alpha_e^{\text{PM}_{2.5-10}}$, and total volume extinction coefficient, α_e , in SI units at Caltech tower. Explain what scattering regime these particles are in and why the extinction coefficient for the smaller particles is larger.

The volume extinction coefficient (α_e , in units of m^{-1}) is calculated from the mass extinction coefficient (k_e) and the mass concentration (ρ) using the formula $\alpha_e = k_e \times \rho$ (**Schneider ch. 5, pg. 143**). First, the given mass extinction coefficients are converted to SI units (m^2kg^{-1}):

- $k_e^{\text{PM2.5}} = 4 \frac{\text{m}^2}{\text{g}} \times \frac{1000 \text{ g}}{1 \text{ kg}} = 4000 \text{ m}^2\text{kg}^{-1}$
- $k_e^{\text{PM2.5-10}} = 1 \frac{\text{m}^2}{\text{g}} \times \frac{1000 \text{ g}}{1 \text{ kg}} = 1000 \text{ m}^2\text{kg}^{-1}$

Using the mass concentrations from **part 1** ($\rho_{\text{PM2.5}} = 1.63 \times 10^{-8} \text{ kg/m}^3$ and $\rho_{\text{PM2.5-10}} = 1.35 \times 10^{-8} \text{ kg/m}^3$), we compute the volume extinction coefficients:

$$\alpha_e^{\text{PM2.5}} = (4000 \text{ m}^2\text{kg}^{-1}) \times (1.63 \times 10^{-8} \text{ kg/m}^3) = 6.52 \times 10^{-5} \text{ m}^{-1}$$

$$\alpha_e^{\text{PM2.5-10}} = (1000 \text{ m}^2\text{kg}^{-1}) \times (1.35 \times 10^{-8} \text{ kg/m}^3) = 1.35 \times 10^{-5} \text{ m}^{-1}$$

The total coefficient is the sum of the components.

$$\alpha_e = \alpha_e^{\text{PM2.5}} + \alpha_e^{\text{PM2.5-10}} = (6.52 \times 10^{-5} \text{ m}^{-1}) + (1.35 \times 10^{-5} \text{ m}^{-1}) = 7.87 \times 10^{-5} \text{ m}^{-1}$$

Thus, the volume extinction coefficients are $\alpha_e^{\text{PM2.5}} = 6.52 \times 10^{-5} \text{ m}^{-1}$ and $\alpha_e^{\text{PM2.5-10}} = 1.35 \times 10^{-5} \text{ m}^{-1}$.

The total volume extinction coefficient is $\alpha_e = 7.87 \times 10^{-5} \text{ m}^{-1}$.

Scattering Regime

The problem concerns the scattering of shortwave (solar) radiation. We can assume a representative wavelength (λ) for visible light, such as $\lambda \approx 0.5 \mu\text{m}$. The particles in question have diameters (d) up to $10 \mu\text{m}$.

Rayleigh scattering applies only when particles are much smaller than the wavelength (e.g., $d \leq 1/10\lambda$) (**Schneider ch. 4, pg. 118**). Our particles are comparable in size to, or larger than, the wavelength of light. Therefore, their interaction with shortwave radiation is described by **Mie scattering** (**Schneider ch. 4, pg. 119**).

Why k_e is Larger for Smaller Particles

The mass extinction coefficient (k_e) is a measure of how effectively a substance scatters or absorbs radiation per unit of mass (**Petty ch. 12, pg. 356**). For a fixed total mass of pollutants (i.e. 1 kg), that mass can be distributed among a few large particles or a great number of small particles.

If the mass is distributed among many smaller particles (like PM2.5), the total geometric cross-sectional area of all those particles combined is much larger than the total area of the fewer, larger particles. Because scattering is proportional to the cross-sectional area of the particles, a cloud of smaller droplets is optically thicker and scatters more light for the same total mass of water (or, in this case, pollutant) (**Schneider ch. 5 pg. 163**). Thus, the PM2.5 particles are more “efficient” at extinction per unit mass, giving them a larger k_e .

Part 3

Assume the planetary boundary layer over LA (the height to which the particulates are mixed) is 1km. Compute the vertical optical depth by integration. Use a scale height of 7km.

The vertical optical depth (τ) is the vertical integral of the volume extinction coefficient ($\alpha_e(z)$) from the surface ($z = 0$) to the top of the atmosphere ($z = \infty$) (**Schneider ch. 5, pg. 148, eq. 5.9**).

The problem states that the particulates are mixed within the 1km planetary boundary layer (PBL). We assume “mixed” implies the particulate mixing ratio is constant within this layer, meaning the mass concentration of the particles follows the same exponential decay as the air density. Above the PBL ($h = 1 \text{ km}$), we assume the particulate concentration is zero.

The volume extinction coefficient at height z is therefore modeled as:

$$\alpha_e(z) = \begin{cases} \alpha_e(0) \exp(-z/H) & \text{for } 0 \leq z \leq h \\ 0 & \text{for } z > h \end{cases}$$

where:

- $\alpha_e(0) = 7.87 \times 10^{-5} \text{ m}^{-1}$ is the total volume extinction coefficient at the surface (from **part 2**)
- $h = 1 \text{ km} = 1000 \text{ m}$ is the height of the PBL
- $H = 7 \text{ km} = 7000 \text{ m}$ is the given scale height

We can now compute the total vertical optical depth by integrating $\alpha_e(z)$ from the surface to the top of the PBL:

$$\begin{aligned}\tau &= \int_0^h \alpha_e(z) dz = \int_0^{1000} (7.87 \times 10^{-5} \exp(-z/7000)) dz \\ &= (7.87 \times 10^{-5}) [-7000 \exp(-z/7000)]_0^{1000} \\ &= (7.87 \times 10^{-5}) \times (-7000) [\exp(-1000/7000) - \exp(0)] \\ &= (7.87 \times 10^{-5}) \times (-7000) [\exp(-1/7) - 1] \\ &\approx 0.07334\end{aligned}$$

The vertical optical depth of the particulates over Pasadena is approximately 0.073.

Part 4

Compute the fractional and absolute reduction in solar radiation from TOA to the surface at solar noon on the day you do this assignment.

Determine Zenith Angle and Slant Path

The amount of atmosphere the light passes through depends on the sun's angle. We know that the solar altitude (angle from the horizon) at solar noon is 39.6° (on November 6, 2025 from <https://www.timeanddate.com/sun/usa/pasadena-ca>). The formulas use the solar zenith angle (θ_z), which is the angle from the vertical:

$$\theta_z = 90^\circ - \text{altitude} = 90^\circ - 39.6^\circ = 50.4^\circ$$

The sunlight is not coming from straight up, so it must pass through more of the pollution layer. We must convert our vertical optical depth ($\hat{\tau}$) from **part 3** into a slant path optical depth (τ_s):

$$\tau_s = \frac{\hat{\tau}}{\cos(\theta_z)} \quad (\text{Schneider ch. 5, pg. 147})$$

Using $\hat{\tau} \approx 0.07334$ from **part 3**:

$$\tau_s = \frac{0.07334}{\cos(50.4^\circ)} \approx 0.115$$

This is the actual optical depth the sunlight must pass through to reach the surface.

Compute the Fractional Reduction

We use the exponential attenuation law (Beer's Law) to find out what fraction of light is blocked (**Schneider ch. 5, pg. 144**). First, we find the transmittance ($\mathcal{T}$), which is the fraction of light that gets through.

$$\mathcal{T} = \exp(-\tau_s) = \exp(-0.1150) \approx 0.8913$$

This means 89.13% of the direct sunlight successfully passes through the pollution. The fractional reduction is the rest of the light: the part that was scattered by the pollution.

$$\text{Fractional Reduction} = 1 - \mathcal{T} = 1 - 0.8914 = 0.1087$$

This means the particulates scatter approximately 10.9% of the incoming direct solar radiation at noon.

Compute the Absolute Reduction

Now we find what 10.9% of the total available solar energy is. First, we need the solar flux at the Top of the Atmosphere (TOA) that is hitting a horizontal surface (the ground) at solar noon. We use the standard solar constant, S_0 .

- Solar Constant, $S_0 \approx 1360 \text{ Wm}^{-2}$ (**Hartmann ch. 3.12, pg. 88**)
- TOA Flux on horizontal surface, $F_{\text{TOA}} = S_0 \times \cos(\theta_z)$

$$F_{\text{TOA}} = (1360 \text{ Wm}^{-2}) \times \cos(50.4^\circ) \approx 866.9 \text{ Wm}^{-2}$$

This is the total direct solar energy that would have hit the surface if there were no pollution. The absolute reduction is 10.9% of this value.

$$\text{Absolute Reduction} = F_{\text{TOA}} \times (\text{Fractional Reduction}) = (866.9 \text{ Wm}^{-2}) \times (0.1087) \approx 94.21 \text{ Wm}^{-2}$$

At solar noon on this day, the particulates reduced the direct solar energy reaching the surface by approximately 94.2 Wm⁻².

2 Heating Rate for Well-Mixed Absorber

Suppose a gas that absorbs solar radiation has a uniform mixing ratio of 1 g kg^{-1} and an absorption cross section of $5 \text{ m}^2 \text{ kg}^{-1}$. Assume an isothermal atmosphere with $T = 260 \text{ K}$, and a surface pressure of 105 Pa .

Part 1

At what altitude is the rate of energy absorption per unit volume maximal when the sun is directly overhead?

The rate of energy absorption per unit volume (i.e. the volumetric heating rate) H (in W/m^3) at an altitude z is given by the convergence of the net downward flux, F^{net} (**Schneider ch. 5, pg. 151**):

$$H(z) = -\frac{dF^{\text{net}}}{dz}$$

For solar radiation, we are concerned with the downward flux $F(z)$. The flux is attenuated from the TOA according to Beer's Law. Since the sun is directly overhead, the solar zenith angle is 0° and the path length is vertical.

$$F(z) = F_{\text{TOA}} \cdot \exp(-\hat{\tau}(z))$$

where $\hat{\tau}(z)$ is the vertical absorption optical depth from the TOA down to altitude z (**Schneider ch. 5, pg. 148**).

$$\hat{\tau}(z) = \int_z^\infty \alpha_a(z') dz'$$

Here, $\alpha_a(z)$ is the volume absorption coefficient. From this definition, $\frac{d\hat{\tau}}{dz} = -\alpha_a(z)$. We can now express the heating rate $H(z)$ by differentiating the flux:

$$H(z) = -\frac{d}{dz} [F_{\text{TOA}} \exp(-\hat{\tau}(z))] = -F_{\text{TOA}} \cdot \exp(-\hat{\tau}(z)) \cdot \left(-\frac{d\hat{\tau}}{dz}\right) = F_{\text{TOA}} \cdot \exp(-\hat{\tau}(z)) \cdot \alpha_a(z)$$

This equation shows the heating rate is the product of the flux reaching a level, $F(z)$, and the local absorptivity, $\alpha_a(z)$. To find the maximum, we must express $\alpha_a(z)$ and $\hat{\tau}(z)$. In an isothermal atmosphere, the air density $\rho_{\text{air}}(z)$ decays exponentially with a constant scale height, H_p :

$$\rho_{\text{air}}(z) = \rho_s \exp(-z/H_p)$$

The volume absorption coefficient $\alpha_a(z)$ is $\alpha_a(z) = \rho_{\text{gas}}(z) \cdot k_a$, where k_a is the mass absorption coefficient ($5 \text{ m}^2 \text{ kg}^{-1}$) and ρ_{gas} is the density of the absorber. Since the gas has a uniform mass mixing ratio $r = 1 \text{ g kg}^{-1} = 0.001$, $\rho_{\text{gas}}(z) = r \cdot \rho_{\text{air}}(z)$.

$$\alpha_a(z) = (r \cdot k_a) \cdot \rho_{\text{air}}(z) = (r \cdot k_a \cdot \rho_s) \cdot \exp(-z/H_p)$$

The optical depth $\hat{\tau}(z)$ is the integral of this:

$$\hat{\tau}(z) = \int_z^\infty (r k_a \rho_s) \exp(-z'/H_p) dz' = (r k_a \rho_s H_p) \cdot \exp(-z/H_p)$$

Comparing our equations for $\alpha_a(z)$ and $\hat{\tau}(z)$, we find a simple relationship:

$$\alpha_a(z) = \frac{\hat{\tau}(z)}{H_p}$$

Substituting this into our heating rate equation:

$$H(z) = F_{\text{TOA}} \cdot \exp(-\hat{\tau}(z)) \cdot \left(\frac{\hat{\tau}(z)}{H_p} \right) = \frac{F_{\text{TOA}}}{H_p} \cdot \hat{\tau}(z) \exp(-\hat{\tau}(z))$$

To find the maximum heating rate, we can find the maximum of the function $f(\hat{\tau}) = \hat{\tau} e^{-\hat{\tau}}$ by taking its derivative with respect to $\hat{\tau}$ and setting it to zero:

$$\frac{df}{d\hat{\tau}} = \frac{d}{d\hat{\tau}}(\hat{\tau} e^{-\hat{\tau}}) = (1 \cdot e^{-\hat{\tau}}) + (\hat{\tau} \cdot (-e^{-\hat{\tau}})) = e^{-\hat{\tau}}(1 - \hat{\tau})$$

$$\frac{df}{d\hat{\tau}} = 0 \implies 1 - \hat{\tau} = 0 \implies \hat{\tau}(z) = 1$$

The maximum rate of absorption occurs at the altitude z where the vertical optical depth from space is equal to 1. We now solve for this altitude z . We know $\hat{\tau}(z) = \hat{\tau}_s \exp(-z/H_p)$, where $\hat{\tau}_s = \hat{\tau}(0)$ is the total optical depth at the surface.

$$1 = \hat{\tau}_s \exp(-z/H_p) \implies \frac{1}{\hat{\tau}_s} = \exp(-z/H_p) \implies z = H_p \ln(\hat{\tau}_s)$$

We must compute H_p and $\hat{\tau}_s$.

Calculate Scale Height (H_p)

For an isothermal atmosphere, $H_p = \frac{R_d T}{g}$. We use the gas constant for dry air, $R_d \approx 287 \text{ J kg}^{-1} \text{ K}^{-1}$, and standard gravity, $g = 9.81 \text{ m s}^{-2}$.

$$H_p = \frac{(287 \text{ J kg}^{-1} \text{ K}^{-1}) \cdot (260 \text{ K})}{9.8 \text{ m s}^{-2}} = 7606.5 \text{ m}$$

Calculate Total Optical Depth ($\hat{\tau}_s$)

$$\hat{\tau}_s = \hat{\tau}(0) = \int_0^\infty \alpha_a(z) dz = (r \cdot k_a) \int_0^\infty \rho_{\text{air}}(z) dz$$

The integral $\int_0^\infty \rho_{\text{air}}(z) dz$ is the total mass of the atmospheric column, M_{col} . From hydrostatic equilibrium, this mass is $M_{\text{col}} = p_s/g$.

$$\hat{\tau}_s = (r \cdot k_a) \cdot \left(\frac{p_s}{g} \right)$$

Using the given values: $r = 1 \text{ g kg}^{-1} = 0.001$, $k_a = 5 \text{ m}^2 \text{ kg}^{-1}$, $p_s = 1 \times 10^5 \text{ Pa}$, $g = 9.81 \text{ m s}^{-2}$

$$\hat{\tau}_s = (0.001) \cdot (5 \text{ m}^2 \text{ kg}^{-1}) \cdot \left(\frac{1 \times 10^5 \text{ Pa}}{9.8 \text{ m s}^{-2}} \right) \approx 50.97$$

Calculate Altitude z

The altitude of maximum absorption is then:

$$z = H_p \ln(\hat{\tau}_s) = (7616.3 \text{ m}) \cdot \ln(51.02) \approx 29\,903 \text{ m} \quad \text{or} \quad 29.9 \text{ km}$$

The rate of energy absorption per unit volume is maximal at an altitude of approximately 29.9 km.

Part 2

At what altitude is the rate of energy absorption per unit volume maximal when the solar zenith angle is 45° ? What difference does the zenith angle make?

A non-zero zenith angle, θ , means the solar radiation travels along a slant path through the atmosphere. In a plane-parallel atmosphere, the slant path length ds is related to the vertical altitude change dz by $ds = -dz/\mu$, where $\mu = \cos(\theta)$ (**Schneider ch. 5, pg. 147**). The vertical optical depth, $\hat{\tau}(z)$, is defined from the top of the atmosphere down. The optical depth along the slant path, $\tau(z)$, is therefore:

$$\tau(z) = \hat{\tau}(z)/\mu$$

The downward vertical flux (energy per horizontal area) at altitude z is the TOA horizontal flux, $F_{\text{TOA, horiz}}$, attenuated by the slant optical depth:

$$F_{\text{down}}(z) = F_{\text{TOA, horiz}} \cdot \exp(-\tau(z)) = (S_0\mu) \cdot \exp(-\hat{\tau}(z)/\mu)$$

(where S_0 is the solar flux normal to the beam at TOA). The volumetric heating rate H_V is the convergence of this flux (note: $H_V = +\frac{dF_{\text{down}}}{dz}$ since F_{down} is positive downward, and convergence heats the layer).

$$H_V(z) = \frac{d}{dz} [(S_0\mu) \cdot \exp(-\hat{\tau}/\mu)]$$

Using the chain rule $\left(\frac{d}{dz} = \frac{d\hat{\tau}}{dz} \frac{d}{d\hat{\tau}}\right)$ and the fact that $\frac{d\hat{\tau}}{dz} = -\alpha_a(z)$ (which we saw in **part 1**):

$$\begin{aligned} H_V(z) &= (S_0\mu) \cdot \left[\frac{d}{d\hat{\tau}} \exp(-\hat{\tau}/\mu) \right] \cdot \frac{d\hat{\tau}}{dz} \\ &= (S_0\mu) \cdot \left[-\frac{1}{\mu} \exp(-\hat{\tau}/\mu) \right] \cdot (-\alpha_a(z)) \\ &= S_0 \cdot \exp(-\hat{\tau}/\mu) \cdot \alpha_a(z) \end{aligned}$$

This equation gives the heating rate. To find its maximum, we substitute $\alpha_a(z) = \hat{\tau}(z)/H_p$ (derived in **part 1**):

$$H_V(z) = S_0 \cdot \exp(-\hat{\tau}/\mu) \cdot \left(\frac{\hat{\tau}}{H_p} \right) = \frac{S_0}{H_p} \cdot \hat{\tau} \exp(-\hat{\tau}/\mu)$$

We find the maximum by differentiating $f(\hat{\tau}) = \hat{\tau} \exp(-\hat{\tau}/\mu)$ with respect to $\hat{\tau}$ and setting it to 0:

$$\frac{df}{d\hat{\tau}} = (1 \cdot \exp(-\hat{\tau}/\mu)) + \left(\hat{\tau} \cdot \left(-\frac{1}{\mu}\right) \exp(-\hat{\tau}/\mu) \right) = \exp(-\hat{\tau}/\mu) \cdot \left(1 - \frac{\hat{\tau}}{\mu} \right)$$

Setting the derivative to 0 gives $1 - \frac{\hat{\tau}}{\mu} = 0$, which means:

$$\hat{\tau}(z) = \mu$$

The maximum absorption occurs where the vertical optical depth, $\hat{\tau}$, equals the cosine of the zenith angle. For this problem, $\theta = 45^\circ$, so:

$$\mu = \cos(45^\circ) \approx 0.707$$

The maximum absorption occurs at the altitude z where $\hat{\tau}(z) = 0.707$. From **part 1**, we know the scale height $H_p = 7606.5 \text{ m}$ and the total vertical optical depth at the surface $\hat{\tau}_s = 50.97$. We solve for z using

$$\hat{\tau}(z) = \hat{\tau}_s \exp(-z/H_p):$$

$$\begin{aligned}\hat{\tau}(z) &= \hat{\tau}_s \exp(-z/H_p) \\ \exp(-z/H_p) &= \frac{\hat{\tau}(z)}{\hat{\tau}_s} \\ -z/H_p &= \ln\left(\frac{\hat{\tau}(z)}{\hat{\tau}_s}\right) \\ z &= -H_p \ln\left(\frac{\hat{\tau}(z)}{\hat{\tau}_s}\right) \\ &= -(7606.5 \text{ m}) \ln\left(\frac{0.707}{50.97}\right) \\ &\approx 32\,539 \text{ m} \quad \text{or} \quad 32.5 \text{ km}\end{aligned}$$

So, the maximum absorption now occurs at **32.5 km**. We see that the zenith angle **increases the altitude of maximum absorption** (from 29.9 km in **part 1** to 32.5 km). This is because the maximum absorption always occurs at the level where the slant optical depth (the optical depth along the beam's path) is equal to 1.

$$\text{Slant Optical Depth } \tau(z) = \frac{\hat{\tau}(z)}{\mu}$$

The level of maximum absorption is where $\tau(z) = 1$, or $\hat{\tau}(z) = \mu$.

- **Part 1 (Overhead Sun):** $\theta = 0^\circ \implies \mu = 1$. The maximum absorption was at $\hat{\tau} = 1$.
- **Part 2 (Slanted Sun):** $\theta = 45^\circ \implies \mu \approx 0.707$. The maximum absorption is at $\hat{\tau} = 0.707$.

Because a smaller vertical optical depth (0.707) is required, this level must be at a **higher altitude**, where the air is less dense. The slanted path means the radiation travels a longer distance to pass through a given vertical layer, so it is attenuated more effectively, and the “sweet spot” of $\tau = 1$ is reached higher up in the atmosphere.

Part 3

If the frequency range over which absorption takes place contains 5% of the total solar energy flux, what is the heating rate (K d^{-1}) at the level of maximum energy absorption? What is the heating rate one density scale height above and below the level of maximum absorption? (Use globally averaged insolation.)

H_T at Level of Maximum Absorption

“Globally averaged insolation” refers to the solar constant $S_0 \approx 1360 \text{ W m}^{-2}$ averaged over the entire Earth's surface, which is $F_{\text{TOA, total}} = S_0/4 \approx 340 \text{ W m}^{-2}$. The problem states that only 5% of this flux is absorbed:

$$F_{\text{abs}} = 0.05 \times F_{\text{TOA, total}} = 0.05 \times 340 \text{ W m}^{-2} = 17 \text{ W m}^{-2}$$

For globally averaged calculations, a standard effective solar zenith angle is $\theta = 60^\circ$, which gives a cosine of $\mu = \cos(60^\circ) = 0.5$.

Volumetric Heating Rate (H_V)

From **part 2**, the volumetric heating rate H_V at a level $\hat{\tau}$ is:

$$H_V(\hat{\tau}) = \frac{F_{\text{abs}}}{\mu H_p} \cdot \hat{\tau} \cdot \exp(-\hat{\tau}/\mu)$$

Also from **part 2**, the maximum absorption occurs at the altitude where $\hat{\tau} = \mu$. Plugging $\hat{\tau} = \mu$ into the equation gives the maximum volumetric heating rate:

$$H_{V,\text{max}} = \frac{F_{\text{abs}}}{\mu H_p} \cdot (\mu) \cdot \exp(-\mu/\mu) = \frac{F_{\text{abs}}}{H_p} e$$

Using $H_p = 7606.5 \text{ m}$ (from **part 1**):

$$H_{V,\text{max}} = \frac{17 \text{ W m}^{-2}}{7606.5 \text{ m} \cdot e} \approx 8.22 \times 10^{-4} \text{ W m}^{-3}$$

Density (ρ)

We need the density at this level of maximum absorption. We can find it using the relation derived in **part 1**, $\rho(z) = \hat{\tau}(z)/(rk_a H_p)$. At our level, $\hat{\tau} = \mu$:

$$\rho_{\max_abs} = \frac{\mu}{rk_a H_p} = \frac{0.5}{(0.001) \cdot (5 \text{ m}^2 \text{ kg}^{-1}) \cdot (7606.5 \text{ m})} \approx 0.013 \text{ 15 kg m}^{-3}$$

Heating Rate (H_T)

Now we convert to K d^{-1} , using $c_p \approx 1005 \text{ J kg}^{-1} \text{ K}^{-1}$, the specific heat capacity of dry air at constant pressure:

$$H_{T,\max}(\text{K/s}) = \frac{H_{V,\max}}{\rho_{\max_abs} \cdot c_p} = \frac{8.22 \times 10^{-4} \text{ W m}^{-3}}{0.013 \text{ 15 kg m}^{-3} \cdot 1005 \text{ J kg}^{-1} \text{ K}^{-1}} \approx 6.22 \times 10^{-5} \text{ K s}^{-1}$$

Converting seconds to days:

$$H_{T,\max}(\text{K/day}) = 6.22 \times 10^{-5} \text{ K s}^{-1} \times 86400 \text{ s d}^{-1} \approx 5.38 \text{ K d}^{-1}$$

So, the heating rate at the level of maximum absorption is 5.38 K d⁻¹.

H_T One Scale Height Above Max Absorption

Let the level of max absorption be z . We are now at $z' = z + H_p$.

- **New Optical Depth ($\hat{\tau}'$):** $\hat{\tau}(z) \propto \exp(-z/H_p)$. So, $\hat{\tau}' = \hat{\tau}(z) \cdot e^{-1} = \mu e^{-1}$
- **New Density (ρ'):** $\rho(z) \propto \exp(-z/H_p)$. So, $\rho' = \rho(z) \cdot e^{-1} = \rho_{\max_abs} \cdot e^{-1}$

Volumetric Heating Rate (H'_V)

$$\begin{aligned} H'_V(\hat{\tau}') &= \frac{F_{\text{abs}}}{\mu H_p} \cdot \hat{\tau}' \cdot \exp\left(-\frac{\hat{\tau}'}{\mu}\right) && \text{[part 2]} \\ &= \frac{F_{\text{abs}}}{\mu H_p} \cdot (\mu e^{-1}) \cdot \exp\left(-\frac{\mu e^{-1}}{\mu}\right) && \text{[new optical depth]} \\ &= \left(\frac{F_{\text{abs}}}{H_p e}\right) \cdot \exp\left(-\frac{1}{e}\right) && \text{[cancel } \mu] \\ &= e^{-1/e} H_{V,\max} && \text{[previous def. of } H_{V,\max}] \end{aligned}$$

Heating Rate (H'_T)

$$\begin{aligned} H'_T &= \frac{H'_V}{\rho' c_p} && \text{[convert heating rate to K d}^{-1}] \\ &= \frac{e^{-1/e} H_{V,\max}}{(\rho_{\max_abs} \cdot e^{-1}) c_p} && \text{[sub. new values]} \\ &= \left(\frac{H_{V,\max}}{\rho_{\max_abs} c_p}\right) \cdot \frac{e^{-1/e}}{e^{-1}} && \text{[isolate constants]} \\ &= H_{T,\max} \cdot \exp(1) \cdot \exp(-1/e) && \text{[prev. def. of } H_{T,\max}] \\ &= H_{T,\max} \cdot \exp(1 - 1/e) && \text{[simplify]} \\ &= (5.38 \text{ K d}^{-1}) \cdot \exp(1 - 1/e) && \text{[plug-in prev. calc. value]} \\ &\approx 10.1 \text{ K d}^{-1} \end{aligned}$$

Thus, the heating rate (in K/day) one scale height above max absorption is higher at 10.1 K d⁻¹ because the density ρ has decreased more (by e^{-1}) than the volumetric heating H_V has (by $\exp(-e^{-1})$).

H_T One Scale Height Below Max Absorption

We are now at $z'' = z - H_p$.

- **New Optical Depth ($\hat{\tau}''$):** $\hat{\tau}'' = \hat{\tau}(z) \cdot e^1 = \mu e$
- **New Density (ρ''):** $\rho'' = \rho(z) \cdot e^1 = \rho_{\max_abs} \cdot e$

Volumetric Heating Rate (H_V'')

$$\begin{aligned}
 H_V(\hat{\tau}'') &= \frac{F_{\text{abs}}}{\mu H_p} \cdot \hat{\tau}'' \cdot \exp(-\hat{\tau}''/\mu) && \text{[part 2]} \\
 &= \frac{F_{\text{abs}}}{\mu H_p} \cdot (\mu e) \cdot \exp(-(\mu e)/\mu) && \text{[new optical depth]} \\
 &= \left(\frac{F_{\text{abs}}}{H_p} \right) \cdot e \cdot \exp(-e) && \text{[cancel } \mu] \\
 &= \left(\frac{F_{\text{abs}}}{H_p e} \right) \cdot e^2 \cdot e^{-e} && \text{[add } e \text{ to sub. } H_{V,\max}] \\
 &= e^{2-e} H_{V,\max} && \text{[prev. def. of } H_{V,\max}]
 \end{aligned}$$

Heating Rate (H_T'')

$$\begin{aligned}
 H_T'' &= \frac{H_V''}{\rho'' c_p} && \text{[convert heating rate to K d}^{-1}] \\
 &= \frac{e^{2-e} H_{V,\max}}{(\rho_{\max_abs} \cdot e) c_p} && \text{[sub. new values]} \\
 &= \left(\frac{H_{V,\max}}{\rho_{\max_abs} c_p} \right) \cdot \frac{e^{2-e}}{e} && \text{[isolate constants]} \\
 &= H_{T,\max} \cdot e^{1-e} && \text{[prev. def. of } H_{T,\max}] \\
 &= (5.38 \text{ K d}^{-1}) \cdot e^{1-e} && \text{[plug-in prev. calc. value]} \\
 &\approx 0.964 \text{ K d}^{-1}
 \end{aligned}$$

Thus, the heating rate (in K/day) one scale height below max absorption is much lower at 0.964 K d⁻¹.

Part 2

Give two examples of well-mixed absorbers and two examples of absorbers that are not well mixed in earth's atmosphere.

An absorber's mixing status generally depends on the lifetime of the gas: gases with long lifetimes (slow sources and sinks) relative to atmospheric mixing times are "well-mixed," while those with short lifetimes (fast sources and sinks) are "not well-mixed."

Well-Mixed Absorbers

Carbon dioxide (CO₂) is well-mixed because its sources and sinks are slow compared to the time it takes for the atmosphere to mix. This results in a mixing ratio that is approximately constant with latitude and altitude up to about 100 km (**Hartmann ch. 3, pg. 78-79**).

Oxygen (O₂) : While its absorption is different from greenhouse gases (i.e. it has rotational transitions via its magnetic dipole (**Schneider ch. 4, pg. 112**)), it is an absorber used for microwave soundings (**Schneider ch. 4, pg. 112; Petty ch. 12, pg. 383**). Its total column content is relatively stable and "varies only by about 5%," indicating it is well-mixed (**Petty ch. 12, pg. 383-384**).

Not Well-Mixed Absorbers

Water vapor (H_2O) is a dominant absorber, but its distribution is “highly variable” (**Hartmann ch. 3, pg. 78**). This is because its sources (evaporation) and sinks (condensation) are fast and depend strongly on temperature, causing its concentration to vary significantly by location and altitude (**Petty ch. 12, pg. 384**).

Ozone's (O_3) concentration is highly variable. It has “fast sources and sinks in the stratosphere” where it is most abundant, and it is also “produced in association with photochemical smog” near the surface (**Hartmann ch. 3, pg. 79**). This results in a concentration profile that is strongly dependent on altitude and location, making it not well-mixed.